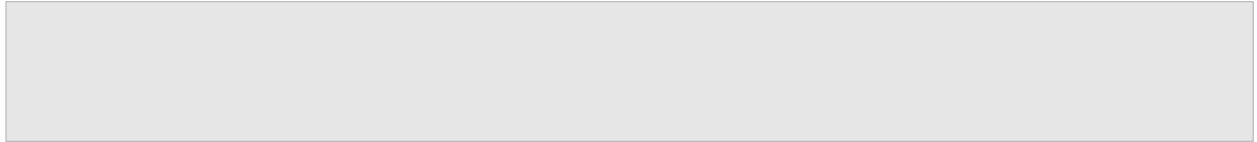


Design and cloning of CRISPR guide RNAs

Risk assessment



Reviewed: May 28, 2025



B >

Making pooled guide RNA libraries

- (1.) For each guide RNA, flank the protospacer with restriction sites that generate overhangs compatible with the vector's U6 cloning site. 

Cas	Plasmid	Format	Enzyme	Guide	Oligo sequence
AsCas12a	pRDA052	Array	BsmBI	23 bp	5'-CGTCTCAgatNNNNN...NNNNNtttttgaatCGAGACG-3'
SpCas9	pX330	sgRNA	BbsI	G + 19 bp	5'-GAAGACTGcaccGNNNN...NNNNNgtttAAGTCTTC-3'

Add one pair of unique primer binding sites to each construct. This allows selective amplification of individual subpools from a combined oligonucleotide pool if desired.

No.	Forward primer (= 5' Flank)	3' Flank	Reverse primer
1	5'-AGGCACTTGCTCGTACGACG-3'	5'-TTAAGGTGCCGGGCCACAT-3'	5'-ATGTGGGCCCGGCACCTTAA-3'
2	5'-GTGTAACCCGTAGGGCACCT-3'	5'-GTCGAAGGACTGCTCTCGAC-3'	5'-GTCGAGAGCAGTCCTTCGAC-3'
3	5'-CAGCGCCAATGGGCTTTCGA-3'	5'-CGACAGGCTCTTAAGCGGCT-3'	5'-AGCCGCTTAAGAGCCTGTGCG-3'
4	5'-CTACAGGTACCGGTCCTGAG-3'	5'-CGGATCGTCACGCTAGGTAC-3'	5'-GTACCTAGCGTGACGATCCG-3'
5	5'-CATGTTGCCCTGAGGCACAG-3'	5'-AGCCTTTCGGGACCTAACGG-3'	5'-CCGTTAGGTCCCGAAAGGCT-3'
6	5'-GGTCGTCGCATCACAATGCG-3'	5'-CGTCACATTGGCGCTCGAGA-3'	5'-TCTCGAGCGCCAATGTGACG-3'

- (2.) Amplify each subpool from approximately 40 ng of oligonucleotide pool using the matching primer pair. Use an annealing temperature of 53 °C and limit the PCR to 20 cycles or less to avoid strand swapping between subpools.
- (3.) Purify the amplicons over a silica membrane.
- (4.) Digest the purified PCR product with the matching Type IIS restriction enzyme under standard conditions. No inactivation is required before ligation.

>>

Ligation into U6-based vectors

- (1.) Linearize 1 µg entry vector with the designated restriction enzyme in 20 µL 1 × restriction digest buffer. 
- (2.) *Optional:* When cloning guide RNA libraries, purify the linearized product from an agarose gel.

